

# Deep Reinforcement Learning

## *Multi-armed bandits*

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# Content

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- Multi-armed bandits
- Action-value methods
- A simple bandit algorithm
- What we have not discussed
- Some next steps

# Where are we?

| Chapter | Topic                                     | Content                          |
|---------|-------------------------------------------|----------------------------------|
|         | Basics & tabular methods                  |                                  |
| 1       | Multi-armed bandits                       | Exploration-exploitation dilemma |
| 2       | Markov decision processes                 |                                  |
| 3       | Dynamic programming                       |                                  |
| 4       | Monte Carlo methods                       |                                  |
| 5       | Temporal difference learning & Q-learning |                                  |
|         | Deep-learning-based methods               |                                  |
|         | Model-Based Control                       |                                  |
|         | Advanced Topics                           |                                  |

*Lecture contents*

# Example: Route planning (OH16 to Hansaplatz by car)

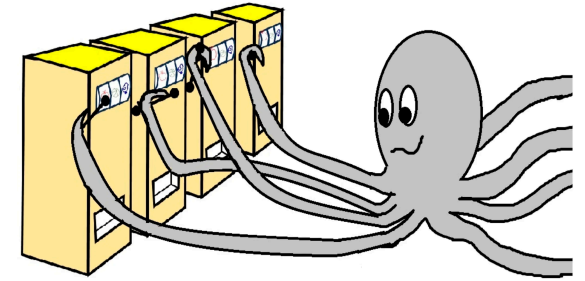
# Multi-armed bandits

# Multi-armed bandits

- Let us assume that we have a slot machine and we repeatedly can choose between  $k$  different actions
- After each choice  $a_t$  you receive a numerical reward  $r_t$  chosen from a stationary probability distribution\*
- Objective: maximize the **expected total reward** over some time period (e.g., over 1000 action selections, or *time steps*)
- We refer to this as the **value**:

$$q(a) = \mathbb{E}[r_t | a_t = a]$$

- If we knew  $q(a)$ , then it would be easy to choose!
- Instead, we have to rely on estimates  $Q_t(a)$  which we can iteratively update based on past experience



A multi-armed bandit (Ferreira, Schubert, and Scotti 2024)

\*We will have a discussion on the notation and random variables in the next lecture (MDPs)

# Action-value methods



# Action-value methods

- *Action-value methods*: estimate the values of actions and then use these estimates to make action selection decisions.
- What was the true value again? → the **expected reward**
- What does that mean for a finite number of samples? → we take the **mean**

$$Q_t(a) = \frac{\sum_{i=1}^{t-1} r_i \cdot \mathbb{1}_{a_i=a}}{\sum_{i=1}^{t-1} \mathbb{1}_{a_i=a}}.$$

Here,  $\mathbb{1}_x$  denotes the random variable that is 1 if the predicate  $x$  is true and 0 if  $x$  is false. (In other words, we're *counting* how often the action  $a$  was taken).

- For  $\sum_{i=1}^{t-1} \mathbb{1}_{a_i=a} = 0$  (we have never used the action  $a$ ), we can choose a default value such as  $Q_t(a) = 0$ .
- We call this the *sample-average method* (Sutton and Barto 1998).

**Question:** How do we pick the action  $a_t$ ?  $\Rightarrow$  just take the max!

$$a_t = \arg \max_a Q_t(a),$$

(with ties broken arbitrarily).

# A simple bandit algorithm

# The $k$ -armed testbed

- Let's study a large number of randomly generated  $k$ -armed bandit problems (here:  $k = 10$ ).
- For each bandit problem the action values  $q(a)$ ,  $a = 1, \dots, 10$ , are selected to a normal distribution:

$$q(a) \sim \mathcal{N}(0, 1).$$

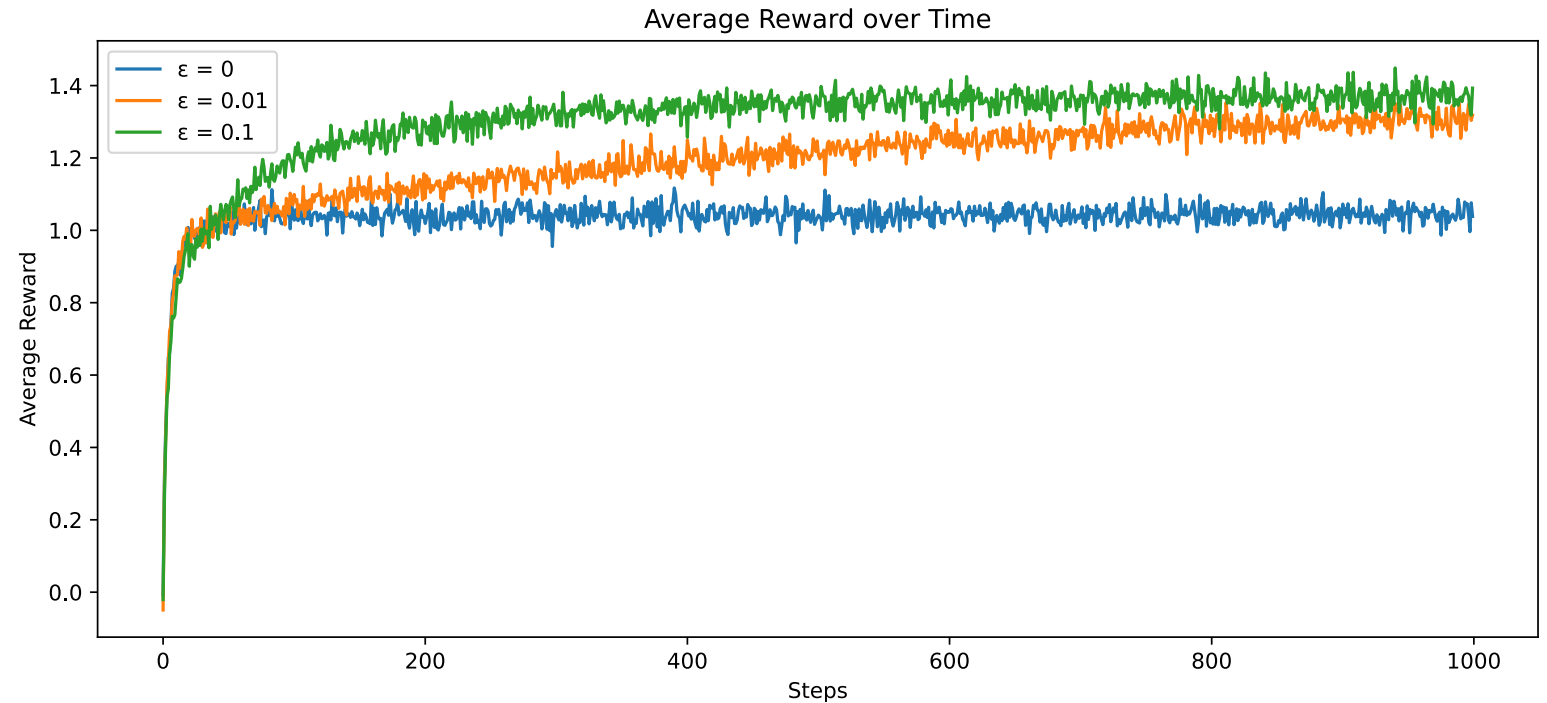
- The corresponding reward is a random variable as well:

$$r_t \sim \mathcal{N}(q(a_t), 1).$$

- One **run**: perform  $T = 1000$  steps. For each time step, report the average reward received until then.
- **Statistics**: repeat the above experiment 2000 times.

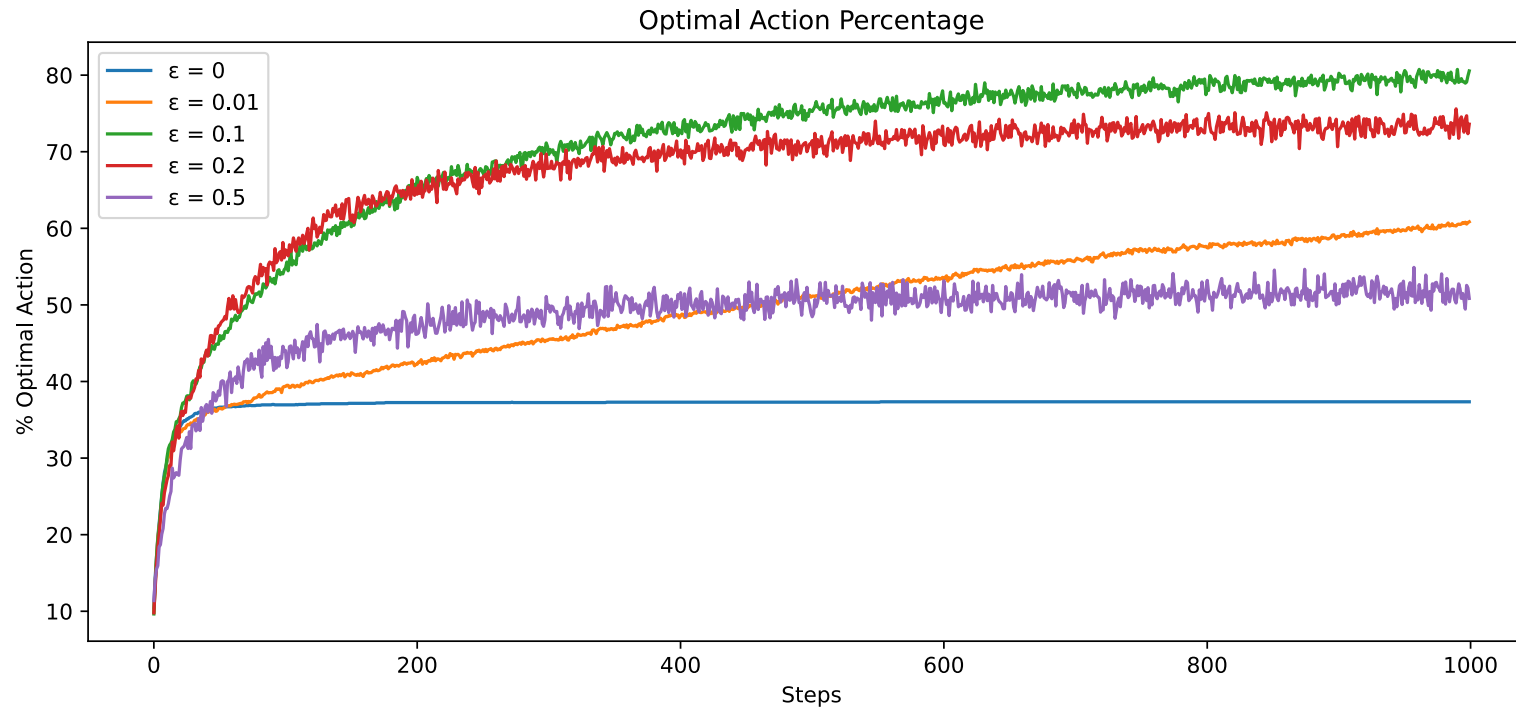
# The $k$ -armed testbed: rewards

- First, we are going to follow a **greedy** strategy: pick the maximizing action *always*.
- Then, we will compare against  $\epsilon$  greedy:
  - pick the maximizing action  $1 - \epsilon$  of the time,
  - pick a random action  $\epsilon$  of the time.



What is  $\epsilon$ 's job?  $\Rightarrow$  the **exploration**

# The $k$ -armed testbed: optimal actions



- Too little (or no) exploration is harmful.
- Too much exploration is also harmful.
- Should we think about a **schedule** in terms of the exploration?

# A simple bandit algorithm

```
# Initialization
for i in range(k):
    Q(a) = 0
    N(a) = 0

# Run forever
while(True):

    # exploration versus exploitation
    if rand() > epsilon:
        a = argmax(Q)      # exploitation
    else:
        a = randint(k)     # exploration

    r = bandit(a)

    # Error correction towards target r
    N(a) = N(a) + 1
    Q(a) = Q(a) + (1/N(a)) * (r - Q(a))
```

*A simple version of the  $\epsilon$  greedy bandit*

## Some next steps

# Computing $Q(a)$ on the fly

Calculating  $Q(a)$  anew every time appears to be expensive. For a single action, consider that we have received a sequence of rewards  $r_1, \dots, r_{t-1}$  so far:

$$Q_t = \frac{r_1 + r_2 + \dots + r_{t-1}}{t - 1}.$$

Wouldn't it be easier if we could just update incrementally?  $\Rightarrow$  let's try

$$\begin{aligned} Q_{t+1} &= \frac{1}{t} \sum_{i=1}^t r_i = \frac{1}{t} \left( r_t + \sum_{i=1}^{t-1} r_i \right) = \frac{1}{t} \left( r_t + (t-1) \frac{1}{t-1} \sum_{i=1}^{t-1} r_i \right) = \frac{1}{t} (r_t + (t-1)Q_t) = \frac{1}{t} (r_t + tQ_t - Q_t) \\ &= Q_t + \frac{1}{t} (r_t - Q_t). \end{aligned}$$

The expression **[target — OldEstimate]** is an **error estimate**, used to steer us closer to the **target**.

We will see a formulae of the type

$$\text{NewEstimate} \leftarrow \text{OldEstimate} + \text{StepSize} [\text{target} - \text{OldEstimate}]$$

frequently from now on!



# Non-stationary problems

- In reinforcement learning, problems are often *non-stationary*, meaning that processes change over time (e.g., due to updates of the **policy**  $\pi$ ).
- Recent rewards should be more relevant than those longer in the past!
- Let's define some constant **step size**  $\alpha \in (0, 1]$ :

$$\begin{aligned} Q_{t+1} &= Q_t + \alpha[r_t - Q_t] = \alpha r_t + (1 - \alpha)Q_t \\ &= \alpha r_t + (1 - \alpha)[\alpha r_{t-1} + (1 - \alpha)Q_{t-1}] = \alpha r_t + (1 - \alpha)\alpha r_{t-1} + (1 - \alpha)^2 Q_{t-1} \\ &= \alpha r_t + (1 - \alpha)\alpha r_{t-1} + (1 - \alpha)^2 \alpha r_{t-2} + \dots + (1 - \alpha)^t r_1 \\ &= (1 - \alpha)^t r_1 + \sum_{i=1}^t \alpha (1 - \alpha)^{t-i} r_i. \end{aligned}$$

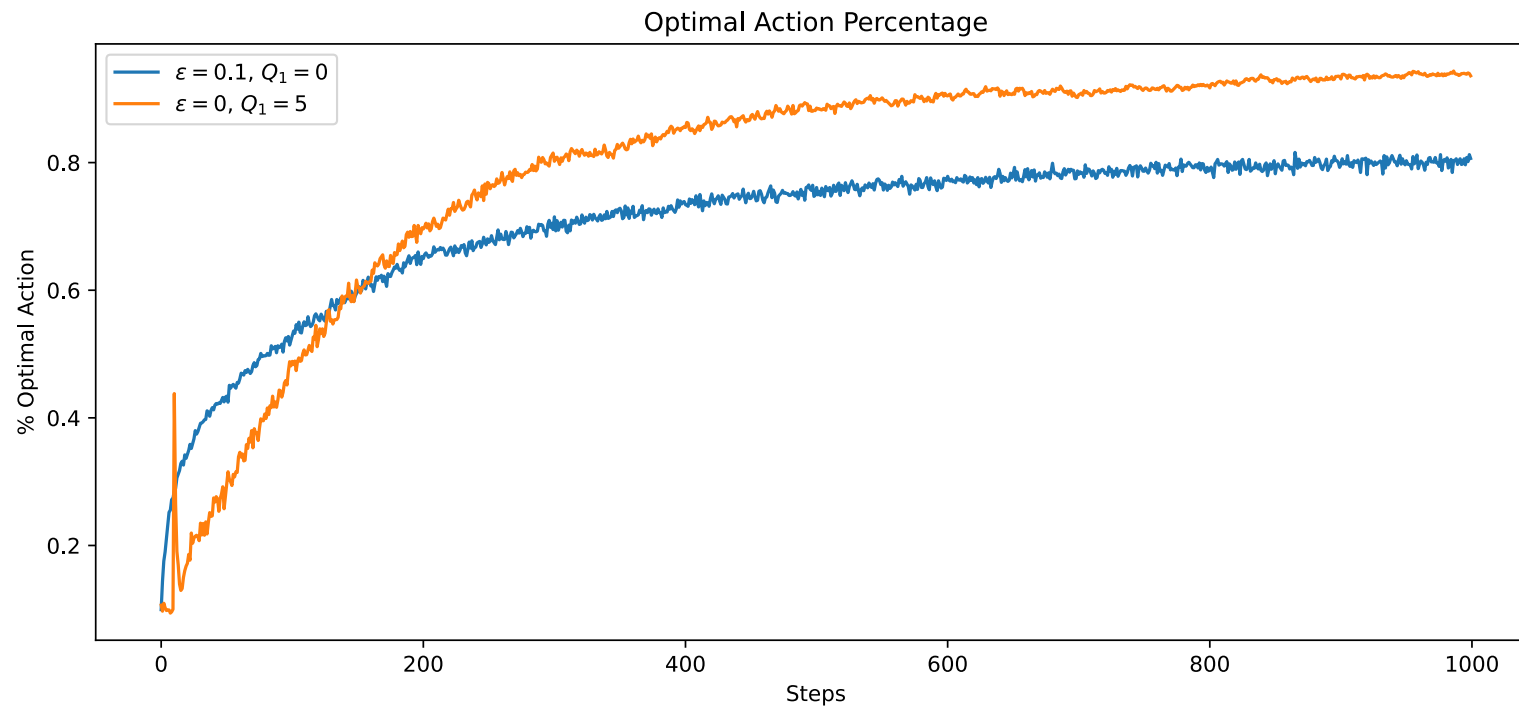
- This is called an (*exponential-recency*) *weighted average*, as

$$(1 - \alpha)^t + \sum_{i=1}^t \alpha (1 - \alpha)^{t-i} = 1.$$

- For time dependent weights  $\alpha_t$ , we require  $\sum_{i=1}^{\infty} \alpha_t(a) = \infty$  and  $\sum_{i=1}^{\infty} \alpha_t^2(a) < \infty$  to ensure convergence. Does this hold for  $\alpha_t = \frac{1}{t}$  and  $\alpha_t = \alpha$ ? (Does it have to hold?)

# Bias in the selection of initial values

- All methods up to now are *biased* in the sense that they depend on the (arbitrarily selected) initial guesses  $Q_1(a)$ .
- Can we use this to our advantage and increase exploration?
- Let us choose overly **optimistic initial values**!
- For  $q(a) \sim \mathcal{N}(1, 0)$ , an initial guess of  $Q_1(a) = 5$  is certainly unrealistically high.



# What we have not discussed

Two topics (and also many more) topics that we have not discussed:

- Bandits with **upper confidence bounds (UCB)**:
  - Along with the estimate for  $Q_t$ , we assign an upper confidence bound based on the number of times  $N_t$  we have selected this action.
  - The larger  $N_t$ , the more confident we are in our estimate.
  - We do not select our next action according to the maximal estimate of  $Q_t$ , but to  $Q_t$  **plus** the confidence bound!
  - The less certain we are, the higher this bound will be  $\Rightarrow$  improved exploration.
- Gradient bandits:
  - We assign a **preference** for each bandit arm  $a$  and introduce the probability  $\pi_t(a)$  for choosing this arm.
  - We can then derive a gradient ascent scheme for the preferences.

# Summary / what we have learned

- Multi-armed bandits model decision making problems where we want to maximize the expected return (the value) of an action in an unknown and uncertain environment.
- Action-value methods estimate the values of actions and then use these estimates to make action selection decisions.
- The exploration-exploitation dilemma describes the challenge of making use of what we know, versus needing to explore to further improve in the long run.
- A simple approach is the  $\epsilon$ -greedy algorithm, where we explore randomly with probability  $\epsilon$ , and otherwise follow the optimal (greedy) action.

# References

- Ferreira, Rafael Pereira, Emil Schubert, and Américo Scotti. 2024. “Exploring Multi-Armed Bandit (MAB) as an AI Tool for Optimising GMA-WAAM Path Planning.” *Journal of Manufacturing and Materials Processing* 8 (3).
- Sutton, R. S., and A. G. Barto. 1998. *Reinforcement Learning: An Introduction*. MIT Press.